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NeuroBridge AI

Wearer-Controlled Multimodal Communication Bridge & Sovereign Safety Net Platform

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TRADITIONAL AAC APPROACH (Deficit & Masking)

- **Unilateral Masking Focus:** Forces autistic youth to mimic neurotypical norms, to leading autistic burnout.
- **Prohibitive Hardware Cost:** Dedicated AAC devices (Dynavox) cost $4,000–$8,000 with long insurance delays.
- **One-Way Static Output:** Rigid icon grids speak words out loud but provide zero subtext or intent translation.
- **Blind to Situational Mutism:** Assumes user can coordinate motor menus during acute sensory shutdowns.
- **Naive AI Gaslighting:** Generic chatbots mistake hostile bullying ('nobody for likes you') friendly idioms.

NEUROBRIDGE DUAL-ENGINE (Double Empathy AI)

- **Reciprocal Subtext Translation:** Translates idioms for autistic wearers and directness for neurotypical peers.
- **Ubiquitous Zero-Cost Hardware:** Operates in browser on existing laptops/smartphones (https://hack2health.vercel.app).
- **4-in-1 Multimodal Timeline:** Unifies Speech, Switch-Scanning, AAC Chips, and 3D MediaPipe Stim Gestures.
- **Wearer-Sovereign Safety Net:** On-demand GPS & Communication Passport dispatch 24/7 without surveillance.
- **Anti-Gaslighting Hostility Shield:** Detects malicious insults and protects against (RSD). Rejection Dysphoria

THE PROBLEM WE SOLVE (The Social Cliff)

- **The 40% Segregation Cliff:** Over 40% of autistic students spend school in classes segregated (NCES Data).
- **Abrupt Support Termination:** Transition to college brings sudden loss of 1:1 aides. paraprofessional
- **Situational Mutism In Crises:** High sensory environments (cafeterias >75dB) cause complete vocal speech loss.
- **Masking & Autistic Burnout:** Forcing neurotypical imitation causes chronic fatigue, anxiety, and depression.
- **Dignified Healthcare Mission:** Empowering autonomous social agency and mutual cross-neurotype respect.

OUR MULTIMODAL ASSISTIVE INTELLIGENCE LOOP

1. **INGEST** — Audio, Switch, AAC, 3D Stims
2. **RADAR** — Acoustic dB & Noise Monitor
3. **KINEMATICS** — 800ms Buffer & Fingertip Curl
4. **DOUBLE EMPATHY** — Milton (2012) Pragmatic Rules
5. **DUAL-ENGINE** — Gemini 2.5 Flash + Offline Regex
6. **CLARIFY** — Subtext Cards & Intent Shield
7. **SAFETY NET** — GPS Beacon & Medical Passport
8. **DELIVER** — SSE Dashboard & Peer Circles

AI ADAPTIVE DUAL-ENGINE: Continuously personalizes vocabulary & suppresses downvoted idioms via real-time feedback

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TECHNICAL APPROACH

SYSTEM ARCHITECTURE & WORKFLOW

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DATA SOURCES

Spoken Audio Stream
Web Speech API streaming raw acoustic conversation in-browser.

Single-Switch Scanning
Accessible scanning keyboard & for fine motor apraxia support.

Typed AAC Symbol Chips
Pre-configured rapid non-verbal communication symbol cards.

3D Hand Stim Tracking
Google MediaPipe 21-joint 3D kinematic landmark vectors.

Sensory Radar Decibels
Web Audio API real-time sound acoustic pressure (SPL) meter.

DATA PROCESSING

Multimodal Ingestion
Unified stream normalization via centralized /api/analyze route.

Kinematic Buffer
12-frame rolling buffer & (~800ms) tracking velocity reversals.

Fingertip Curl Check
Biometric angle verification eliminates resting hand false alarms.

Disfluency Cleansing
Speech filler removal extracts core emotional sentiment accurately.

Session State Cache
Client cache persists gestures calibrated & emergency profile.

PREDICTION ENGINE

Google Gemini 2.5 Flash
Primary LLM: deep pragmatic & subtext resolution reasoning.

Offline Regex Engine
Zero-downtime heuristic for & fallback transit tunnels basements.

Double Empathy Rules
Bidirectional classifier: and idiom, sarcasm, bluntness, intent.

Anti-Gaslighting Shield
Hostility detection: warns of bullying, refuses to excuse abuse.

Adaptive Feedback Loop
Downvoting idioms personalizes clarification frequency live.

SOVEREIGN OUTPUTS

Subtext Clarification
Real-time contextual meaning in cards rendered conversation.

Sovereign Safety Net
Instant GPS coordinates & alert situational mutism dispatch.

Communication Passport
Digital medical ID for first responders, teachers, and peers.

Tactful Exit Cards
One-tap pre-rehearsed boundary cards preventing burnout.

Peer Circles & Wingman
Parallel-play virtual study rooms with zero small-talk demands.

TECHNOLOGIES USED

Frontend Application
React 19, Vite 8.3, Custom CSS Glassmorphism Design System.

Backend Architecture
Node.js, Express.js REST API, Server-Sent Events (SSE).

AI & Computer Vision
Google Gemini 2.5 Flash, Vision MediaPipe WebAssembly.

Assistive APIs
Web Speech API, DeviceMotion API (G-force), Geolocation API.

Deployment & DevOps
Vercel Edge Serverless, GitHub CI/CD, 18 Automated Self-Tests.

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FEASIBILITY & VIABILITY

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FEASIBILITY

• Zero Hardware Barrier

Runs on ubiquitous web browsers (Chrome, Safari, Edge) on existing student phones and laptops. Eliminates $7,000 proprietary hardware barriers completely.

• Production Proven Today

Live full-stack deployment active on Vercel at with https://hack2health.vercel.app instant edge execution across worldwide edge nodes.

• Cost & Scale Efficiency

Serverless micro-architecture executes Gemini 2.5 only Flash on demand, costing <$0.01 per conversation with zero idle infrastructure cost.

KEY CHALLENGES

• High Acoustic Noise

Cafeteria background chatter (75–85 dB) degrades standard browser microphone speech recognition during accuracy social transitions.

• Mobile OS Sensor Throttling

Mobile Android and iOS operating systems suspend tab background browser sensors and camera feeds when the screen is locked.

• AAC Typo & Lexical Noise

Autistic disfluency, dyspraxia, or rapid typing errors generates phonetic spelling ('neeeed a brek') that bypass simple regex matching.

OUR STRATEGY

• Multimodal Input Fallback

When noise peaks on the Sensory Radar, the wearer to seamlessly shifts single-switch scanning or one-tap quick-response AAC chips with zero downtime.

• Native Daemon Packaging

Phase 3 roadmap packages the web app into a Native an Capacitor/React daemon with Android Foreground Service for locked-screen fall monitoring.

• LLM Intent Reconstruction

Gemini 2.5 Flash semantically extracts the core intent emotional/functional from disjointed spelling without judging grammar or penalizing the user.

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IMPACT AND BENEFITS

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POTENTIAL IMPACT ON TARGET AUDIENCE

AUTISTIC YOUTH (AGES 14-24)

Overcoming The Social Cliff

- Eliminates reliance on 1:1 human aides upon high school graduation.
- Prevents situational mutism panic via one-tap AAC and s tim gestures.
- Tactful Exit Cards lower autistic masking burnout and s ocial anxiety.
- Empowers genuine self-advocacy and dignified boundary s etting.

PARENTS & GUARDIANS

Sovereign Peace of Mind

- Real-time GPS safe perimeter telemetry streaming on Gua rdian View.
- Immediate automated alert dispatch during acute sensory overload crises.
- Zero invasive 24/7 spying: telemetry transmitted only b y wearer consent.
- Direct emergency phone dialer with verified coordinates & sound levels.

EDUCATORS & PEERS

Mutual Understanding (Double Empathy)

- Educates neurotypical peers: directness is constructive honesty, not rudeness.
- Reframes hand stimming as joyous self-regulation rather than a crisis.
- Equips campus accessibility offices with Digital Commun ication Passports.
- Fosters inclusive parallel-play study and co-working en vironments.

BENEFITS OF THE SOLUTION

SOCIAL BENEFITS

Inclusion Without Forced Masking

- Fosters authentic neurodivergent friendships without so cial camouflage.
- Social Wingman matches intense shared passions (keyboar ds, transit).
- Replaces social ambiguity with predictable, structured interactions.
- Destigmatizes somatic stimming and AAC communication in public spaces.

ECONOMIC BENEFITS

Democratizing $7,000 Medical Tech

- Saves families $4,000–$8,000 per user vs proprietary de dicated hardware.
- Reduces institutional special education paraprofessiona l staffing costs.
- Zero subscription paywall: free open-access browser pro totype.
- Runs on existing student hardware without insurance pre -approvals.

HEALTHCARE & CLINICAL BENEFITS

Preventative Mental Health

- Drastically reduces chronic autistic burnout, anxiety, and depression.
- Reduces hospitalizations caused by acute sensory panic disorientations.
- Equips first responders with vital non-verbal medical c risis profiles.
- 18 automated self-tests guarantee verified API and syst em robustness.

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RESEARCH AND REFERENCES

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PEER-REVIEWED CLINICAL & SCIENTIFIC LITERATURE

1. The Double Empathy Problem (Milton, 2012)

Milton, D. E. (2012). 'On the ontological status of autism: the double empathy problem.' Disability & Society, 27(6), 883-887.

Link: https://doi.org/10.1080/09687599.2012.710008

Healthcare Impact: Theoretical core: Proves cross-neurotype miscommunication is a reciprocal two-way mismatch, not an autistic deficit to be masked.

2. Educational Segregation & The Social Cliff (NCES, 2024)

U.S. Dept. of Education, National Center for Education Statistics. Digest of Education Statistics, Table 204.60 & IDEA Section 618.

Link: https://nces.ed.gov/programs/digest/d23/tables/dt23_204.60.asp

Healthcare Impact: Sourced demographic data: Over 40% of autistic students are segregated in K-12 schooling, facing the abrupt Social Cliff upon transition.

3. Defining Autistic Burnout & Camouflaging (Raymaker et al., 2020)

Raymaker, D. M., et al. (2020). 'Having All of Your Internal Resources Exhausted: Defining Autistic Burnout.' Autism in Adulthood, 2(2), 132-143.

Link: https://doi.org/10.1089/aut.2019.0079

Healthcare Impact: Clinical foundation for Tactful Exit Cards: Demonstrates that continuous masking leads to severe cognitive and functional exhaustion.

4. Rejection Sensitive Dysphoria & Neurodivergence

Bedrossian, L. (2021). 'Understand and address rejection sensitive dysphoria.' Mental Health Practice, 24(6).

Link: https://doi.org/10.7748/mhp.2021.e1564

Healthcare Impact: Directly guides our Anti-Gaslighting Hostility Shield: Protects wearers from misinterpreting abusive insults as benign idioms.

LIVE PLATFORM, OPEN-SOURCE CODE & SYSTEM ASSETS

- **Live Deployed Application**

  https://hack2health.vercel.app

  Active production deployment with Google Gemini 2.5 Flash LLM integration.

- **Mobile Guardian Safety Companion**

  https://hack2health.vercel.app/?view=guardian

  Live real-time telemetry dashboard streaming GPS coordinates & decibel readings.

- **Official GitHub Repository**

  https://github.com/tharunya0105-afk/hack2health

  Full open-source codebase with React 19 frontend, Express backend & gesture modules.

- **Automated End-to-End Test Suite**

  backend/test-suite.js

  18 automated tests verifying multimodal ingestion, scanning keyboards & adversarial speech.

- **Complete Video Walkthrough**

  gallery/neurobridge_demo_video.mp4

  1.63-minute 30FPS MP4 video walkthrough covering all 15 system capabilities.

- **Google MediaPipe Vision Tasks**

  https://developers.google.com/mediapipe/solutions/vision/hand_landmarker

  Float16 WebAssembly vision model tracking 21 3D landmarks in real time.